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Let the ground move alone: how base isolation protects a stone building

The free guide to the reel on the Salt Lake City and County Building: why a 1894 sandstone building sits on rubber and steel pads, how isolators work, what they cost the building in movement, and a ruler-and-rubber-band experiment.



A stone building that floats

The Salt Lake City and County Building was built between 1891 and 1894 in Richardsonian Romanesque style, with heavy sandstone walls. Stone is strong when it is pressed, but weak when it is shaken sideways, so an earthquake is its enemy.

In the renovation that reopened the building in 1989, engineers lifted it onto a foundation of steel and rubber, a technique called base isolation.

How an isolator works

An isolator is a stack of rubber layers glued between steel plates, sometimes with a lead core. It is very stiff vertically, so it carries the building's weight without sinking. But it is very limber horizontally, so the ground can slide sideways under the building.

Because the system is so flexible sideways, the building swings slowly, and a slowly swinging building feels much smaller earthquake forces. Engineering sources for this project describe over four hundred isolators, which let the building move up to about two feet in any direction.

Built examples

Salt Lake City and County Building, Utah (1894, isolated 1989)	A 5.7-magnitude earthquake in March 2020 closed it briefly; repairs were minimal and it reopened in November 2021.
Lead-rubber bearing	Invented by Dr William Robinson in 1974: rubber layers with a lead core that also soaks up energy.
Fixed-base building	The usual alternative: bolted to the ground, it is built strong and ductile enough to survive moving with the ground.

Try it yourself: slide the ground

1. Stack three books on a smooth tray: this is your stone building.
2. Shake the tray sideways, fast, and watch the books topple or slide together.

3 Now put a layer of small rubber bands, or a sheet of foam, between the tray and the books, loose enough to flex sideways.

4 Shake again at the same speed. The books move much less, because the pads soak up the movement.

5 Notice the price: the tray moves a lot, while the books barely do. A real isolated building needs room to move around it.

The rule to remember

You can fight a force by making a structure stronger, or you can change how much of it reaches the structure.

Default answer	Make it stiffer and stronger.
Form-finder answer	Let the ground move and keep the building out of the movement.
Check	Ask: where can I put a soft layer so the load never arrives?

Go further

Seismic base isolation (Wikipedia)
https://en.wikipedia.org/wiki/Seismic_base_isolation

Salt Lake City and County Building (Wikipedia)
https://en.wikipedia.org/wiki/Salt_Lake_City_and_County_Building

Forell/Elsesser: base isolation
https://forell.com/practice/expertise/base-isolation

Credits
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Facts: Wikipedia (Salt Lake City and County Building, Seismic base isolation), Forell/Elsesser, Structure magazine.

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